

Invertible Linear Transformations

Recall from Section 2.1 that matrix multiplication corresponds to composition of linear transformations. When a matrix A is invertible, the equation $A^{-1}Ax = x$ can be viewed as a statement about linear transformations.

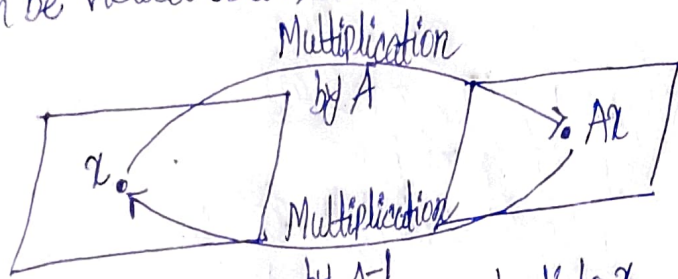


Figure 2 A^{-1} transforms Ax back to x .

A linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is said to be invertible if there exists a function $S: \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that

$$\begin{aligned} S(T(x)) &= x \text{ for all } x \text{ in } \mathbb{R}^n & (1) \\ T(S(x)) &= x \text{ for all } x \text{ in } \mathbb{R}^n & (2) \end{aligned}$$

The next theorem shows that if such an S exists, it is unique and must be a linear transformation. We call S the inverse of T and write it as T^{-1} .

Theorem 9: Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation and let A be the standard matrix for T . Then T is invertible if and only if A is an invertible matrix. In that case, the linear transformation S given by $S(x) = A^{-1}x$ is the unique function satisfying equations (1) and (2).

Proof: By Theorem 10 from Chapter 1, $T(x) = Ax$ for each x in $\mathbb{R}^n$ and A is the unique such matrix. $A = [T(e_1) \ T(e_2) \ \dots \ T(e_n)]$

Let T be invertible. ~~That is, verify whether $S(x)$ exists~~
 Let b be any arbitrary vector in $\mathbb{R}^n$, and $x = S(b)$.
 By equation 2, $T(x) = T(S(b)) = b$, so each b is in the range of T .
 By Defn, T is onto. By Invertible Matrix Theorem, A is invertible, (statement (i)).